Viridis Research — certification record

Run `runs/2026-11-14_P5` 2026-11-14 project P5, task P5-T002 engine `70b3b094b085`
GitHub run: `https://github.com/jdhart81/viridis-engine/actions/runs/43533409125`

A Correction Note on an Early Intelligence Bound Source: The Unrestricted Axiom Is False as Written

Justin Hart, Viridis LLC

Labels awarded by the verifier

Claim	Label	Claim text (as submitted)
C1	**PROVEN**	formal [P5.UnrestrictedBoundFalse]: It is not the case that for all reals dIdt, P, D, kB, T with kB>0, T>0, P>=0, 0<=D<=1 one has dIdt <= maxBitRate P kB T * D, where maxBitRate P kB T = P/(kB*T*log 2); counterinstance dIdt=1, P=D=0, kB=T=1. The historical unrestricted Intelligence Bound axiom is false as written.
C2	**ASSUMED**	conjecture: For cap P D C = P*D/C with C>0, P>0, 0<dD<D: P*D/(D-dD) is the unique power restoring capacity after the loss; under a ceiling P<=Pmax it is attainable iff dD <= D*(1-P/Pmax); it diverges as dD -> D from below; and at D=0 no power restores a positive capacity (sealed statement P5.CompensationUnderCeiling, to be proved in P5-T003).

Sealed statement P5.UnrestrictedBoundFalse

File `lean/ViridisStatements/P5/UnrestrictedBoundFalse.lean`, sha256 `3b83eb37905660e3db1cc68c54e476f4ac5c450b245cec4675195f9cc2e0a95d`, sealed by decision D-019. Axioms used by the proof: Classical.choice, Quot.sound, propext. Candidate sha256 `3f0110cc26b85d009881440a831c81d1ba092cd48135db5ca43685c1e855ba24`. Listing shown with symbol substitution; the exact bytes are in the repository.

```
import Mathlib

/-!
# P5 · The unrestricted Intelligence Bound postulate is false as written

‘Gaia_Intelligence_Theorem.lean‘ (January 2026, SHA-256 ‘37c017145dcf...2098c4‘) declares a
postulate ‘intelligence_bound‘ quantifying the rate ‘dI_dt‘ over *all* reals, with no hypothesis
tying it to any system. This module restates the two historical definitions over Mathlib only
(same algebra, nothing imported or copied from the historical file) and states that the
universally quantified bound is false.

Definitions (mirroring the historical ones):
* ‘landauerEnergy kB T = kB · T · log 2‘;
* ‘maxBitRate P kB T = P / landauerEnergy kB T‘.

‘Statement‘: it is not the case that for all reals ‘dIdt P D kB T‘ with
‘kB > 0‘, ‘T > 0‘, ‘P ≥ 0‘, ‘0 ≤ D ≤ 1‘ we have ‘dIdt ≤ maxBitRate P kB T · D‘.

A witness (not part of the statement) is ‘dIdt = 1‘, ‘P = 0‘, ‘D = 0‘, ‘kB = T = 1‘.

The statement concerns that exact formal text. It says nothing about the Intelligence Bound as a
physical conjecture, which needs a model of the system whose rate is bounded, and nothing about
whether any agent preserves biodiversity.
-/

noncomputable section

namespace ViridisStatements.P5.UnrestrictedBoundFalse

/-- Landauer energy per bit, ‘kB · T · ln 2‘ (mirrors the historical ‘landauer_energy‘). -/
def landauerEnergy (kB T : ℝ) : ℝ := kB * T * Real.log 2

/-- Maximal bit rate at power ‘P‘, ‘P / (kB · T · ln 2)‘ (mirrors the historical
‘max_bit_rate‘). -/
def maxBitRate (P kB T : ℝ) : ℝ := P / landauerEnergy kB T

/-- **Proposed sealed target ‘P5.UnrestrictedBoundFalse‘.** The historical postulate, with the rate
quantified over all reals, is false. -/
def Statement : Prop :=
  ¬ (∀ dIdt P D kB T : ℝ,
      kB > 0 → T > 0 → P ≥ 0 → D ≥ 0 → D ≤ 1 →
      dIdt ≤ maxBitRate P kB T * D)
```

```
/-! ## Sanity checks -/

/-- At zero power the maximal bit rate is zero. -/
example : maxBitRate 0 1 1 = 0 := by
  simp [maxBitRate]

/-- The Landauer energy at `kB = T = 1` is `ln 2 > 0`. -/
example : 0 < landauerEnergy 1 1 := by
  simp [landauerEnergy, Real.log_pos one_lt_two]

/-- The witness `dIdt = 1, P = 0, D = 0, kB = T = 1` meets every guard and violates the
conclusion: the bound evaluates to `0` and `1 ≤ 0` fails. -/
example : (1 : ℝ) > 0 ∧ (1 : ℝ) > 0 ∧ (0 : ℝ) ≥ 0 ∧ (0 : ℝ) ≥ 0 ∧ (0 : ℝ) ≤ 1 ∧
    ¬ ((1 : ℝ) ≤ maxBitRate 0 1 1 * 0) := by
  norm_num

/-- The guards and the arithmetic core of the witness, checked by `decide` on natural numbers:
every guard holds and the conclusion `1 ≤ 0 · 0` fails. -/
example : (0 < 1 ∧ 0 < 1 ∧ 0 ≤ 0 ∧ 0 ≤ 1) ∧ ¬ (1 ≤ 0 * 0) := by
  decide

/-- Negative control: once the rate is tied to the bound by hypothesis, the implication is
trivially true. The missing ingredient in the historical postulate is exactly such a condition. -/
example : ∀ dIdt P D kB T : ℝ,
    kB > 0 → T > 0 → P ≥ 0 → D ≥ 0 → D ≤ 1 →
    dIdt ≤ maxBitRate P kB T * D → dIdt ≤ maxBitRate P kB T * D :=
  fun _ _ _ _ _ _ _ _ _ _ h => h

/-- Negative control: a nonpositive rate never violates the bound when `P, D ≥ 0` and
`kB, T > 0`; so a counterexample needs a positive rate exceeding the capacity. -/
example : ∀ dIdt P D kB T : ℝ,
    kB > 0 → T > 0 → P ≥ 0 → D ≥ 0 → dIdt ≤ 0 →
    dIdt ≤ maxBitRate P kB T * D := by
  intro dIdt P D kB T hkB hT hP hD hr
  have hL : 0 < landauerEnergy kB T := by
    unfold landauerEnergy
    exact mul_pos (mul_pos hkB hT) (Real.log_pos one_lt_two)
  have : 0 ≤ maxBitRate P kB T * D :=
    mul_nonneg (div_nonneg hP hL.le) hD
  linarith

end ViridisStatements.P5.UnrestrictedBoundFalse
```

Other checks

- Lean toolchain leanprover/lean4:v4.28.0, Mathlib 8f9d9cff6bd7; kernel replay by `leanchecker`.
- References: 3 entries, 3 DOIs, 3 resolved, 0 unresolved, 0 unchecked.
- Agent paper: compiled.
- Statement proposals: 0.

How to read this record

ASSUMED: not verified (the default). **PROVEN**: a Lean 4 proof of a statement sealed before the attempt, compiled with the pinned toolchain, matched definitionally to the sealed statement, using only the axioms propext, Classical.choice and Quot.sound, and replayed through the kernel. **TESTED**: the verifier re-ran the claim's analysis script in a sandbox with no network, on input files pinned by hash, and reproduced the cited result files; this does not certify where the data came from or the inference drawn from it. **REPLICATED**: not yet enabled. **ENDORSED/ADOPTED**: granted by people. Novelty is never a label. **TRIVIAL** marks a sealed statement that one of the engine's probe tactics already closes.

This paper was drafted by an AI research agent (Claude, by Anthropic) running the Viridis research engine; formal proof search may use Aristotle (Harmonic). Justin Hart directs the research program. The labels on this page were assigned by an automated verifier, not by the agent, and only they carry certification. Claims in the paper that are not listed here as PROVEN or TESTED are ASSUMED.

Reproduce everything from `REPRODUCE.md` in the run folder; the run's files are listed with hashes in `MANIFEST.sha256`.

A Correction Note on an Early Intelligence Bound Source: The Unrestricted Axiom Is False as Written

Justin Hart
Viridis LLC

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Abstract

An early Viridis Lean source of January 2026 declares an axiom, `intelligence_bound`, stating $\dot{I} \leq \frac{P}{k_B T \ln 2} D$ for every real number $\dot{I}$, under the guards $k_B > 0$, $T > 0$, $P \geq 0$ and $0 \leq D \leq 1$, with no hypothesis linking $\dot{I}$ to any system. The instance $\dot{I} = 1$, $P = D = 0$, $k_B = T = 1$ meets every guard and reduces the conclusion to $1 \leq 0$. We give a Lean 4 proof, against the sealed statement `P5.UnrestrictedBoundFalse`, that the universally quantified proposition is false, and a companion Lean lemma showing that, taken as a hypothesis, it proves every proposition. An environment that assumes the axiom is therefore inconsistent. According to a reading of the corpus, no declaration in the reviewed sources depended on it, so no recorded result is affected. We describe what a well-posed successor needs: a predicate that ties the rate to a modelled physical system, and separate empirical validation. This note is about one piece of formal text. It says nothing about the Intelligence Bound as a physical conjecture. A second correction, on finite compensation of a loss of D, is restated here as a conjecture pending its own proof.

1 Introduction

A machine-checked development is only as reliable as the match between its formal text and what is said about it. A Lean `axiom` is not checked at all: the kernel accepts it as given. If the axiom is false, every theorem proved in its presence is vacuous, because a false hypothesis proves anything. A review of the Viridis research corpus on 11 November 2026 found two early Lean declarations whose formal text diverges from their surrounding prose. Project P5 checks both with the Lean kernel. The first session (P5-T001) proposed exact statements, and these were then sealed. This note covers the first finding: the historical `intelligence_bound` axiom is false as written.

The question matters for record keeping. The axiom was presented as a formal version of a physical bound on information processing. As written, though, it is a claim about all real numbers, and it fails for elementary reasons. Correcting it is not new mathematics. The aim is to keep false text from being cited as a foundation, and to state what a correct successor would have to contain.

2 Background and related work

Landauer's principle [2] associates a minimal dissipation of $k_B T \ln 2$ with each logically irreversible bit operation at temperature T. Dividing a power budget P by this energy gives an upper bound on the rate of irreversible bit operations. The early sources define, over the reals,

$$\text{landauerEnergy}(k_B, T) = k_B T \ln 2, \qquad \text{maxBitRate}(P, k_B, T) = \frac{P}{k_B T \ln 2},$$

and multiply the result by a diversity factor $D \in [0, 1]$. We use Lean 4 [1] and Mathlib [3]. The historical file `Gaia_Intelligence_Theorem.lean` is identified by its SHA-256 prefix `37c017145dcf...2098c4`. It is quoted, not imported: the sealed statement restates the two definitions over Mathlib alone.

3 The historical axiom

The project README quotes the declaration as follows:

```
axiom intelligence_bound :
  ∀(dI_dt P D k_B T : ℝ),
  k_B > 0 →T > 0 →P ≥0 →D ≥0 →D ≤1 →
  dI_dt ≤max_bit_rate P k_B T * D
```

The rate `dI_dt` is a bound variable that ranges over every real number. Nothing in the statement says whose rate it is. The axiom therefore asserts that every real number lies below $\mathrm{maxBitRate}(P, k_B, T) \cdot D$, which cannot hold, because the right-hand side is a fixed real number once the other variables are fixed.

4 Result

Theorem 1 (sealed statement `P5.UnrestrictedBoundFalse`)**.** *It is not the case that for all real $\dot{I}, P, D, k_B, T$ with $k_B > 0$, $T > 0$, $P \geq 0$, $D \geq 0$ and $D \leq 1$ we have $\dot{I} \leq \mathrm{maxBitRate}(P, k_B, T) \cdot D$.*

Counterinstance. Take $\dot{I} = 1$, $P = 0$, $D = 0$, $k_B = T = 1$. Every guard holds. The right-hand side is $\frac{0}{\ln 2} \cdot 0 = 0$, so the conclusion would be $1 \leq 0$, which is false. □

We give a Lean proof of Theorem 1. The automated checker run during the session passed all five stages, and the only axioms used are `propext`, `Classical.choice` and `Quot.sound`. The candidate file is:

```
lemma bound_zero_power (D kB T : ℝ) : maxBitRate 0 kB T * D = 0 := by
  simp [maxBitRate]

theorem main : ViridisStatements.P5.UnrestrictedBoundFalse.Statement := by
  intro h
  -- Counterinstance: dIdt = 1, P = 0, D = 0, kB = T = 1.
  have h1 := h 1 0 0 1 1 one_pos one_pos le_rfl le_rfl zero_le_one
  rw [bound_zero_power] at h1
  exact absurd h1 (by norm_num)

theorem bound_proves_anything
    (h : ∀dIdt P D kB T : ℝ, kB > 0 →T > 0 →P ≥0 →D ≥0 →D ≤1 →
      dIdt ≤maxBitRate P kB T * D)
    (p : Prop) : p :=
  absurd h main
```

Remark 1 (The witness is not degenerate). The witness uses $P = 0$ and $D = 0$, but the falsity does not depend on these boundary values. For any admissible P, D, k_B, T, the rate $\dot{I} = \mathrm{maxBitRate}(P, k_B, T) \cdot D + 1$ violates the bound. In particular this works for $P > 0$ and $D = 1$. The witness also does not rely on Lean's convention $x/0 = 0$: here the denominator is $\ln 2 \neq 0$.

Consequence: inconsistency. The lemma `bound_proves_anything` makes the consequence explicit. If the universally quantified bound is available as a hypothesis, or as an axiom, any proposition p follows, including p and $\neg p$ alike (*ex falso quodlibet*). A Lean environment that declares `intelligence_bound` as an axiom is therefore inconsistent. Every statement can be proved in it, and a successful compilation in that environment certifies nothing.

Impact on the corpus. The 2026-11-11 review reports that no declaration in `Gaia_Intelligence_Theorem.lean` uses the axiom and that no other file in the reviewed corpus imports it. If that is right, no recorded theorem inherits the inconsistency. This rests on reading the sources. It is not established by the Lean proof above, which concerns only the restated proposition.

5 What a well-posed successor needs

The negative control in the sealed module shows what is missing. Once the rate is tied to the bound by hypothesis, the implication holds trivially. A meaningful successor therefore cannot quantify the rate freely. It needs:

1. *A predicate tying the rate to a physical system.* For example, a structure modelling a system with power P at temperature T, whose information rate is *defined* from that model, perhaps as a count of irreversible bit operations per unit time. The bound then becomes a theorem about that structure, derived from an explicit Landauer-type hypothesis, rather than an axiom about arbitrary reals. The role of D also has to be defined inside the model, not just multiplied in.

2. *Separate empirical validation.* Whether such a model describes real agents or ecosystems is an empirical question. No Lean proof answers it. Its premises must be stated and tested against data, and they must be labelled as assumptions until that is done.

Neither item is attempted here.

6 The second correction (pending)

For the record, the second P5 finding is restated without change. Let $\mathrm{cap}(P, D, C) = PD/C$. The historical theorem `no_finite_compensation` actually proves that the finite power $PD/(D - \Delta D)$ restores the capacity after a partial loss $0 < \Delta D < D$.

Conjecture 1 (sealed statement `P5.CompensationUnderCeiling`)**.** *Let $C > 0$ and $P > 0$. (1) If $0 < \Delta D < D$, then $P' = PD/(D - \Delta D)$ is the unique power with $\mathrm{cap}(P', D - \Delta D, C) = \mathrm{cap}(P, D, C)$. (2) If moreover $P \leq P_{\max}$, then such a $P' \leq P_{\max}$ exists if and only if $\Delta D \leq D(1 - P/P_{\max})$. (3) For $D > 0$, $PD/(D - \Delta D) \to \infty$ as $\Delta D \to D^-$. (4) For $D > 0$ and every P', $\mathrm{cap}(P', 0, C) = 0 < \mathrm{cap}(P, D, C)$.*

The proof is task P5-T003. No proof of it is claimed in this note.

7 Limitations

This is a correction of one piece of formal text. It is elementary, and it is not a result about the Intelligence Bound conjecture. Showing that the unrestricted text is false says nothing about

whether a correctly posed bound holds for any physical system. It also says nothing about whether any agent would preserve biodiversity. The statement that no corpus declaration depended on the axiom comes from reading the sources and is not machine-checked. The sealed statement restates the historical definitions instead of importing the historical file. Its fidelity to the original therefore rests on the quoted text and the file hash.

8 Reproducibility

The proof is `proofs/P5.UnrestrictedBoundFalse.lean`. It imports only Mathlib and the sealed module `ViridisStatements.P5.UnrestrictedBoundFalse` (SHA-256 `3b83eb37905660e3db1cc68c54e476f4ac5c`
It compiles with toolchain `leanprover/lean4:v4.28.0` and Mathlib commit `8f9d9cff6bd728b17a24e163c940277`
and it can be checked with `viridis-lean-check proofs/P5.UnrestrictedBoundFalse.lean P5.UnrestrictedB`
No data or external code is used.

9 Changelog

- 2026-11-13 (P5-T001): first version. Proposed both statements with sanity checks and negative controls, and noted the conditions the negative controls need.
- 2026-11-14 (P5-T002): refocused the note on the first correction. Added a Lean proof of the sealed `P5.UnrestrictedBoundFalse` and an explosion lemma, explained the inconsistency and its impact on the corpus, and set out the requirements for a well-posed successor. The second correction remains a conjecture.

References

[1] Leonardo de Moura and Sebastian Ullrich. The Lean 4 theorem prover and programming language. In *Automated Deduction – CADE 28*, volume 12699 of *Lecture Notes in Computer Science*, pages 625–635. Springer, 2021. doi: 10.1007/978-3-030-79876-5_37.

[2] Rolf Landauer. Irreversibility and heat generation in the computing process. *IBM Journal of Research and Development*, 5(3):183–191, 1961. doi: 10.1147/rd.53.0183.

[3] The mathlib Community. The Lean mathematical library. In *Proceedings of the 9th ACM SIGPLAN International Conference on Certified Programs and Proofs (CPP 2020)*, pages 367–381. Association for Computing Machinery, 2020. doi: 10.1145/3372885.3373824.